Cafeteria Waste Detection with Specialized YOLO11 Models and Selective Prediction

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Abstract. Sorting cafeteria waste requires more than recognizing discarded objects: a carton, wrapper, or food container may still require information about its material, contents, or local collection rules. We evaluate a vision-assisted prototype consisting of four specialized YOLO11 detectors for seven cafeteria-item classes and a rule-based policy that either suggests a disposal stream or requests manual review. On 248 previously examined development photographs containing 924 AI-assisted object annotations, the frozen system achieved 87.78% mAP50 and 67.24% mAP50–95. At a confidence threshold of 0.50, object-detection precision was 93.42%, with 78.35% recall. Class-specific thresholds selected using 186 validation photographs raised precision to 97.03% on validation and 97.22% on the historically exposed 248-image cohort; recall fell to 71.43% and 68.18%, respectively. Only six of those 248 images came from the participating school, and a provenance audit found no confirmed independent test cohort or human-verified disposal-bin labels. The results therefore characterize development-set recognition and its precision–recall trade-off, not the accuracy of actual composting or recycling decisions.

Keywords: cafeteria waste, object detection, YOLO11, selective prediction, precision–recall trade-off, evaluation provenance.

1 Introduction

At the end of a school lunch period, a single cafeteria tray may contain food scraps, disposable utensils, a drink carton, and plastic packaging. Although these items are visually distinguishable, they do not necessarily belong in the same disposal stream. An item's correct destination depends not only on what it looks like, but also on its material, contamination, remaining contents, and local collection requirements. The U.S. Environmental Protection Agency advises residents to verify local recycling rules and to empty and clean recyclable containers [1]. A camera may therefore recognize an object correctly without determining how it should be discarded.

Computer vision offers a way to identify several discarded items within a single image, even when they occur together rather than against an isolated background. Object detection is especially relevant because it produces separate locations and classes for the visible items. However, the predictions of a detection model are not equivalent to verified recycling or composting decisions. A useful assistance system must distinguish between confident object recognition, uncertain predictions that warrant human review, and objects that the model fails to detect.

This study investigates that distinction through a prototype containing four specialized YOLO11 detectors and a rule-based disposal-suggestion layer. The detectors cover seven cafeteria-item classes, while the decision layer recommends composting or recycling for selected recognized categories and otherwise defers to manual review. We evaluate the frozen detectors on the available development images, examine whether class-specific confidence thresholds can reduce false-positive detections without retraining, and audit the provenance and limitations of the resulting measurements. The objective is to characterize what the existing recognition system can support, rather than to claim that its object predictions have already been validated as correct disposal decisions.

1.1 Related Work

Early research on automated waste recognition primarily approached the problem as image classification. Thung and Yang developed TrashNet to classify photographs of individual objects into six material-based categories, including glass, metal, paper, plastic, cardboard, and trash [2]. Their work demonstrated the potential of visual waste classification, but the images feature isolated objects against relatively simple backgrounds. Single, Iranmanesh, and Raad later introduced RealWaste, a dataset captured at an operating landfill, and examined how models trained on cleaner material imagery performed when confronted with more realistic waste conditions [3]. Together, these studies show that the appearance and collection context of waste matter when evaluating whether an image classifier can generalize beyond its source dataset.

Other work has shifted from whole-image classification to locating individual objects in more complex scenes. Proença and Simões introduced TACO, a dataset with instance annotations for litter photographed in its surrounding environment, and evaluated instance segmentation with Mask R-CNN [4]. Bashkirova and colleagues

further developed ZeroWaste, a dataset of highly cluttered recycling-conveyor scenes containing deformable and overlapping materials, where conventional detection and segmentation methods face substantial challenges [5]. These studies motivate object-level localization for mixed waste scenes. Nevertheless, material categories, litter-instance masks, and cafeteria disposal eligibility are different targets: even an accurately localized object may require information that is not visible in a photograph before it can be assigned to a local bin.

The object-detection approach used here follows the YOLO family, initially introduced by Redmon and colleagues as a unified framework for predicting object locations and categories from an image [6]. Standardized evaluation practices for object recognition were developed through benchmarks such as Microsoft COCO, which uses localization-sensitive average precision to assess detection quality [7]. The present system uses YOLO11 as implemented by Ultralytics [8]. These tools measure how well objects are recognized and localized, but average precision over a range of detection scores does not describe the precision or coverage of a specific operating threshold. For a decision-assistance prototype, that fixed-threshold behavior also matters.

Research on selective classification offers a framework for systems that withhold uncertain predictions. Geifman and El-Yaniv investigated rejection based on classifier confidence [9] and subsequently developed SelectiveNet, which learns a rejection option jointly with the underlying prediction task [10]. The current study does not implement or claim to outperform either approach; instead, it applies the simpler idea of confidence-based selection to already trained detectors, comparing a shared cutoff against class-specific cutoffs. This distinction is important because higher precision among accepted detections can come at the expense of missed objects and lower coverage.

Finally, evaluation design determines what those measurements can establish. The WILDS benchmark documents how performance can deteriorate when systems are tested across different sources and environments [11]. Kapoor and Narayanan similarly show that data leakage and evaluation design flaws can produce overly optimistic scientific conclusions [12]. These concerns are particularly relevant when waste images come from multiple public sources, model histories are not fully reconstructed, and nominal test images have been examined during development. Accordingly, this study separates development-set recognition results from the independent field evaluation still required to establish reliable cafeteria sorting performance.

2 Data and Annotation Provenance

Public source manifests, source URLs and historical license declarations are preserved externally in `data/datasets.csv` and `data/manifests/`. School originals are preserved separately with SHA-256 identities, conversion records and capture metadata. These records require source/attribution verification before public data redistribution; this work did not independently audit licenses.

The working combined manifest is external `data/combined-seven-school-oct29-20261031/manifest.json`. Its nominal train, validation and test roles do not establish independence for the inherited active checkpoints. The train partition contains 2,052 real photos and 320 additional augmentation views, for 2,372 views. Twenty school training originals contribute 340 views including originals. Additional views do not add independent photographs. Validation contains 186 real photos (seven school); test contains 248 (six school). These are available working data, not a claim that each active specialist was trained from scratch on exactly this single manifest.

The historical test cohort has seven recorded source groups: 73 `school_lunch`, 66 `tetra_pak`, 48 `trash_detection`, 31 `plate`, 20 `utensil`, six `school`, and four `macworx` photographs (248 total). Thus only **6/248 photographs (2.42%)** originate from the participating school submission. The source name `school_lunch` denotes a separate dataset, not additional participating-school captures. The object-detection benchmark consequently cannot be interpreted as an estimate of on-site cafeteria performance.

Evaluation uses the frozen, embedded repaired annotations in `artifacts/validation-label-audit-20261030/audited-eli` and `artifacts/test-label-audit-20261031/audited-all248.json`. The test manifest has 924 object boxes. The working dataset's older test rows are superseded for scoring; do not use its older object totals. The validation manifest has corrections absent from 109 referenced disk label files, including ten with bottle-label differences. The files retain their recorded hashes. Embedded frozen rows, not those older disk labels, define this evaluation. No labels were changed during this finalization.

Annotations are AI-assisted and retain uncertainty; historical words such as "manual" or "verified" are not proof of human authorship. The school review CSV has 123 blank `human_reviewed` entries. Food annotations aim at visually distinct servings, but dense buffet examples expose possible serving-granularity ambiguity. Only **two** hinged-container validation objects and **four** historical-test objects are available. High scores for this class therefore provide very limited evidence.

Table 1. Annotation support by class. Training counts are augmented *views*, not independent photographs; validation and historical test counts use repaired object labels.

Class	Training-view boxes	Validation boxes	Test boxes
food	2501	343	433
plate	722	110	123
utensil	457	101	98
hinged_container	185	2	4
drink_carton	972	94	134
wrapper	316	85	75
bottle	293	42	57

3 Model Architecture and Inference

The system consists of four separately adapted YOLO11 specialist models rather than one newly trained seven-class network. To keep this analysis reproducible, the checkpoint hashes, class mappings, inference settings, decision rules, and benchmark files were recorded in research-freeze.json at commit `defcc6156a4425daf1ae530757634cc541f3c267`. Protected file hashes were checked before and after the analysis. The freeze identifies the configuration evaluated in this paper, although it is not a complete inventory of every historical experiment.

Table 2. Frozen specialist checkpoints, parameter counts, class ownership, and per-model inference resolution.

Specialist	YOLO11 variant	Parameters	Input size	Owned classes
food-container-carton	yolo11n.yaml	2,590,425	640	food, hinged_container, drink_carton
plate-wrapper	yolo11n.yaml	2,605,126	640	plate, wrapper
utensil	yolo11x.yaml	56,877,241	800	utensil
bottle	yolo11s.yaml	9,428,179	640	bottle

Inference uses Ultralytics 8.4.175, PyTorch 2.14.1, Python 3.12, batch 1, single-pass inference, NMS IoU 0.70, `max_det=300` per specialist, no test-time augmentation and `half=False`. The last argument is deprecated by the installed library but was retained to reproduce the frozen protocol. Box coordinates are normalized XYXY. Native per-specialist NMS is followed by concatenation of the owned classes; no additional cross-specialist deduplication is applied.

The experiments were run in a recorded environment using an Apple M1 MacBook Air with 8 GiB of memory and MPS acceleration. These specifications document the computational setting; inference time, energy use, and throughput were not benchmarked. The four checkpoints also have different training and selection histories, so the system should not be described as the product of one controlled training run.

The prediction receipt saves detections down to confidence 0.001 for AP analysis. Display/sorting emits scores at least 0.25. Scores below 0.50 require review when displayed; higher scores use the unchanged policy:

Recognized class	Suggested decision at confidence >=0.50
Food	Compost
Bottle, drink carton	Recycling, without checking emptiness
Plate, utensil, wrapper, hinged container	Review

The policy is owner-selected, not evidence that filled containers or every detected food/package qualifies for the school's real collection stream. Independent bin labels must follow actual acceptance evidence rather than the detector mapping.

4 Evaluation Methods

Detection AP uses the existing native Ultralytics evaluator, with mAP50 and the mean over IoU 0.50:0.05:0.95. Fixed-threshold object precision/recall/F1 use confidence-descending one-to-one IoU>=0.50 matches requiring an exact class. These matching conventions and AP's score sweep are distinct measurements.

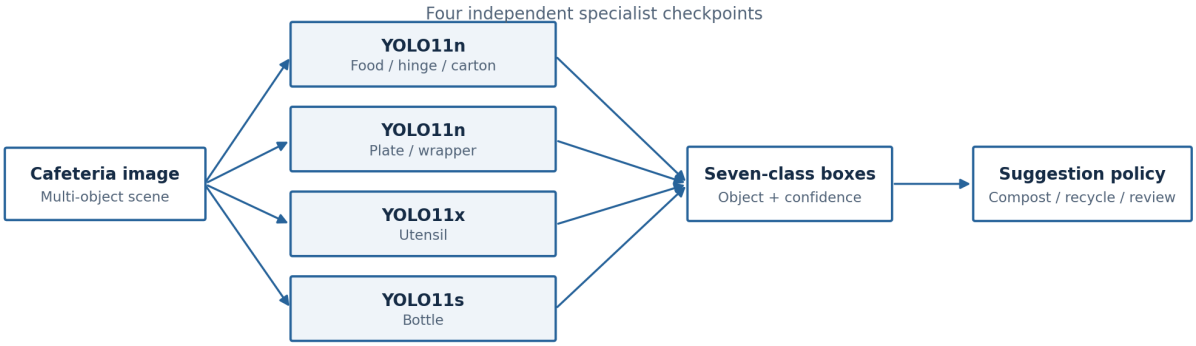


Figure 1: System diagram: four YOLO11 specialists detect seven cafeteria classes before rule-based suggestions or review.

The automatic object proxy is reported two ways. Class-specific matching sums food, bottle and carton matches separately. Joint geometry-first matching orders all displayed detections by confidence and assigns each to at most one unused label by IoU>=0.50 without consulting class/category; exact object class is checked after assignment. This exposes competition between overlapping food/packaging labels. Neither method establishes disposal eligibility.

For an eventual independently labeled bin evaluation, geometry matching must remain independent of the desired bin. For each true bin b, precision is TP_b/(TP_b+FP_b), recall is TP_b/(TP_b+FN_b), and F1 is $2TP_b/(2$TP_b+FP_b+FN_b). Wrong-bin and unmatched automatic predictions contribute false positives; reviews and missed true items remain false negatives for their true bin. Undefined denominators must be reported as undefined, not zero accuracy. Report both correct-bin accuracy among matched automatic decisions and the coverage-sensitive fraction correct automatic/all labeled items. The former excludes reviews, misses and unmatched predictions and must not appear alone. Also report precision across all automatic predictions, object coverage, matched review, missed objects, and unmatched predictions. This prevents abstention or false-positive filtering from inflating apparent performance.

For a predicted box and ground-truth box, intersection-over-union (IoU) is the intersection area divided by the union area. At a fixed minimum IoU of 0.50, sorted, confidence-descending, class-consistent one-to-one matches yield micro precision, recall, and F1:

$$P = \frac{TP}{TP + FP}, \qquad R = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2PR}{P + R}.$$

Because earlier model-selection decisions involved the 248-photo historical cohort, comparisons on those images must be treated as development evidence rather than independent test performance. This distinction follows established concerns about data leakage and test-set reuse in machine-learning research [12]. The baseline AP computation uses the saved native prediction ranks down to confidence 0.001; threshold tuning changes only which detections are accepted at the operating point, not the frozen checkpoint or baseline AP.

5 Confidence Threshold Selection

We first examined whether a single confidence threshold could reduce false detections across all seven classes. Thresholds from 0.50 to 0.99, in increments of 0.01, and an additional threshold of 0.995 were tested on the 186-image validation set. No global threshold met all three requirements: at least 97% micro precision, at least 60% recall, and at least 100 accepted predictions. The best global threshold satisfying the recall and prediction-count requirements was 0.79, yielding 96.04% precision and 68.73% recall.

We therefore considered a separate threshold for each class. For every class, we enumerated the true and false detections accepted at candidate thresholds on validation. A dynamic program combined these class-level possibilities and selected the policy retaining the most correctly detected objects while meeting the same precision, recall, and prediction-count requirements:

$$\max_{\tau} TP(\tau) \quad \text{subject to} \quad P(\tau) \geq 0.97, \quad R(\tau) \geq 0.60, \quad N_{\text{pred}} \geq 100.$$

Table 3. Confidence thresholds selected exclusively from development validation.

Food	Plate	Utensil	Hinged container	Drink carton	Wrapper	Bottle
0.79	**0.83**	**0.50**	**0.50**	**0.84**	**0.83**	**0.91**

The selected thresholds were then applied unchanged to the 248-image historical cohort. They were not retuned on those images or activated in the default sorting pipeline. Because the historical cohort had already influenced earlier development decisions, its scores remain development evidence rather than an independent estimate of field performance.

6 Results

Table 4. Per-class fixed-threshold object-recognition metrics on the exposed 248-photo cohort (confidence>=0.50, IoU>=0.50); AP uses the full frozen score range. These are descriptive development results, not independent bin classification.

Class	TP / FP / FN	Precision	Recall	F1	AP50	AP50–95
food	327 / 10 / 106	97.03%	75.52%	84.94%	83.91%	55.09%
plate	103 / 15 / 20	87.29%	83.74%	85.48%	87.04%	71.60%
utensil	70 / 7 / 28	90.91%	71.43%	80.00%	86.06%	61.26%
hinged_container	4 / 1 / 0	80.00%	100.00%	88.89%	99.50%	91.17%
drink_carton	129 / 1 / 5	99.23%	96.27%	97.73%	97.55%	74.28%
wrapper	58 / 7 / 17	89.23%	77.33%	82.86%	85.13%	57.64%
bottle	33 / 10 / 24	76.74%	57.89%	66.00%	75.27%	59.62%

The macro mAP50/mAP50–95 is **87.78%/67.24%** on the 248-image development cohort and **86.36%/65.25%** on the 186-image frozen validation cohort. All-class test micro precision/recall/F1 at 0.50 is **93.42%/78.35%/85.23%**. Per-class validation metrics are preserved in `results/sorting-verification.json` and the evaluation report.

Table 5. Automatic-suggestion object proxies, explicitly separate from bin accuracy. Recall uses the 624 food/bottle/carton object labels.

Matching convention	Correct object matches	Precision	Recall	F1
Class-specific	489 of 510 automatic suggestions	95.88%	78.37%	86.24%
Joint geometry-first	483 of 510 automatic suggestions	94.71%	77.40%	85.19%

The six joint-matching conflicts are five food boxes and one bottle box assigned to overlapping wrapper labels. They are object/matching conflicts, not independently proven wrong-bin outcomes. The original 248-image replay was previously verified, and a later fresh inference exactly reproduced its predictions. This finalization reuses those receipts; it does not rerun YOLO or train another candidate.

Table 6. Sorting output and coverage denominators on development images.

Quantity	Count / denominator	Value
Compost suggestions	337 / 881 emitted detections	38.25%
Recycling suggestions	173 / 881	19.64%

Quantity	Count / denominator	Value
Automatic suggestion rate	510 / 881	57.89%
Manual-review rate among emitted detections	371 / 881	42.11%
Geometry-matched automatic coverage of labeled objects	489 / 924	52.92%
Exact-class correct automatic matches / labeled objects	483 / 924	52.27%
Geometry-matched review / labeled objects	286 / 924	30.95%
Missed objects	149 / 924	16.13%
Unmatched automatic predictions	21 / 510 automatic suggestions	4.12%
Unmatched review predictions	85 / 371 review predictions	22.91%

No automatic landfill suggestion is produced. The emitted-prediction review rate does not include undetected real items. The 52.27% fraction measures exact-class automatic matches, not correct disposal. No conventional image- or object-iid confidence interval is supplied: objects are clustered within scenes/capture groups, annotations are uncertain, and checkpoint/test exposure is present. A narrow interval on these data would not establish independent generalization.

6.1 Precision–Recall Trade-off After Threshold Selection

Table 7. Overall seven-class micro object recognition at box IoU ≥ 0.50; same source images and labels within each cohort.

Cohort / policy	TP	FP	FN	Precision	Recall	F1
Validation (186), uniform 0.50	638	48	139	**93.00%**	82.11%	87.22%
Validation (186), classwise selected	555	17	222	**97.03%**	71.43%	82.28%
Historical (248), uniform 0.50	724	51	200	**93.42%**	78.35%	85.23%
Historical (248), classwise fixed	630	18	294	**97.22%**	68.18%	80.15%

On the historical cohort, class-specific thresholds reduced false detections from 51 to 18, but they also removed 94 correct detections: true positives fell from 724 to 630. The total number of accepted predictions declined from 775 to 648. Precision increased by 3.80 percentage points, while recall declined by 10.17 percentage points. This change reflects a stricter decision policy rather than an improvement in model weights or mAP. The historical cohort was not used for another round of threshold selection.

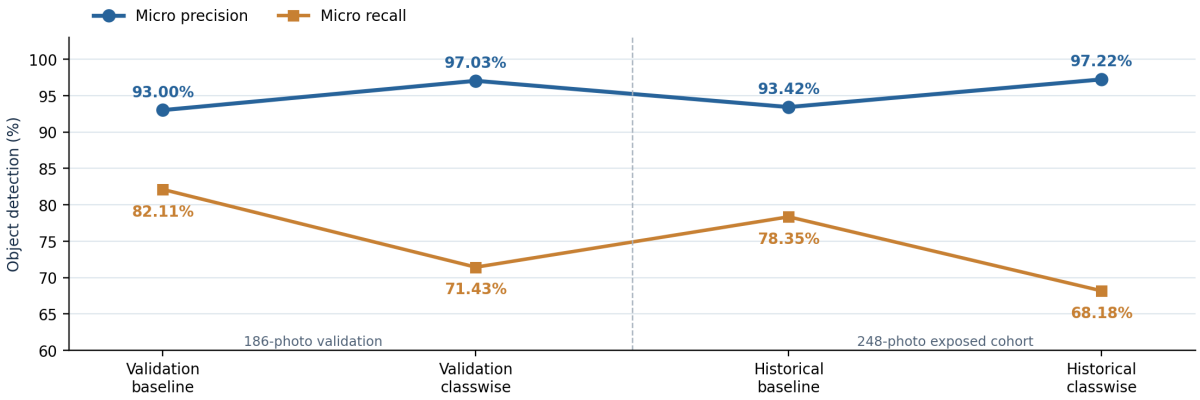


Figure 2: Precision–recall trade-off for default and validation-selected classwise confidence thresholds, measured on exposed development cohorts.

7 Independence and Annotation Audit

To investigate potential overlap and dependence among evaluation images, we audited the local data inventory rather than assuming that a folder labeled `test` was independent. Similar threats from dataset provenance and distribution shift have been documented in the broader machine-learning literature [11, 12]. The inventory contains 102,806 readable image paths (67,422 distinct inodes), which collapse to 14,379 byte-unique images. Aliases and repeated copies are not independent samples. The original/candidate pool contains 9,468 byte identities from raw/processed/public candidate sources and the direct school originals and JPEG conversion directories. Review figures and experiment charts are excluded from this candidate pool.

SHA-256 checks exact bytes; an EXIF-oriented RGB pixel hash checks identical decoded pixels. Eight rotation/reflection versions of a 64-bit grayscale dHash screen near-duplicates against 4,751 images with training-directory or prepared-manifest membership, at Hamming radius five. These are review flags, not proof that two photos are identical. Directory membership is potential exposure; it is not proof that a particular active checkpoint's optimizer consumed every listed image. Capture-group and recorded inference evidence are separately propagated.

The audit parsed 2,398 JSON files. The additional label search examined 207 CSV, 34 XML and 263 YAML files. XML objects were labeled `Plastic Bottle`, not disposal bins; no per-object bin-label format was found in the JSON search. Historical review/verified columns describe object review and do not establish independent human bin truth. No verified human disposal-label set was available for scoring.

Table 8. Candidate provenance status. These categories do not assert that no independent photo exists anywhere; they state what can be established locally.

Status	Byte-unique candidate identities
Exposure/group/duplicate flag	3,842
Unresolved historical provenance	5,482
Original HEIC visual decoding unavailable	144
Confirmed independent	**0**

Reason counts overlap: 3,042 candidates share training-pool bytes/membership, 2,590 share a training-pool group, 2,682 have recorded development inference, 2,707 share a development group, and 588 trigger perceptual training-pool flags. These are not additive counts of optimizer exposures. The HEIC files were byte hashed; their 144 recorded converted school photos were decoded and all share training/development capture groups. Conversion provenance is recorded, rather than independently proved by a HEIC decoder in this environment.

All 123 school photos supplied in the earlier submission and all 21 in the newer submission share training or development capture groups. The newer submission was captured on October 29; four photos were admitted to training and its whole day was excluded from independent testing. For the earlier submission, initial "sealed test" or "no inference" flags are superseded by subsequent model-use ledgers and prediction receipts. Historical test participation in release decisions makes those data development evidence. A nominal split name cannot restore

independence.

The audit also records 42 unreadable/dangling image aliases, one malformed JSON process record, and four inaccessible historical JSON files. It did not modify or repair any original. Unmatched public source photos remain unresolved because complete lifetime fitting/selection histories and all capture relationships are not available. dHash cannot rule out every crop, viewpoint change or adjacent frame. Generic-pretraining exposure is also not certified absent.

Table 9. Independent real-world disposal metrics — unavailable, not zero.

Metric	Independent result	Missing evidence
Precision, recall, F1 by actual bin	Not measured	Human per-object bin labels and independent images
Correct-bin accuracy	Not measured	Independently established disposal eligibility
Automatic coverage of independent objects	Not measured	Independently annotated object inventory
Manual-review rate on independent cohort	Not measured	A confirmed independent cohort run

8 Error Analysis and External-Image Evaluation

The diagnostic decomposition uses frozen object labels and predictions, without editing annotations or changing thresholds. A false-negative label is classified as below threshold if a same-class box has IoU>=0.50 but score<0.50; a localization candidate has same-class IoU>=0.10 but <0.50; otherwise it has no same-class overlap at that cutoff and saved confidence floor. These are competing hypotheses, not human-confirmed causes. A box can overlap other objects or depend on uncertain serving definitions. Background-or-missing-annotation likewise does not distinguish model hallucination from an incomplete label set.

Diagnostic category	Food	Bottle
Correct class matches at automatic threshold	327	33
FN: below automatic threshold	53	19
FN: localization candidate	48	1
FN: no same-class overlap	5	4
FP: localization candidate	2	0
FP: duplicate/competing detection	1	0
FP: background or missing annotation	7	9
FP: other-class overlap/annotation ambiguity	0	1

Errors were concentrated in a small number of difficult scenes. One buffet photograph accounted for 52 of the 106 missed food objects (49.06%), while a single outdoor-trash photograph accounted for 10 of the 24 missed bottles (41.67%). Multiple errors in one scene should not be interpreted as independent failures across many photographs. Both scenes remained in the reported results.

AI visual inspection covered 12 deterministically selected error cases from 10 unique photographs; it is not a random audit or a human-label correction. Examples suggest a reusable-looking sports bottle outside the disposable-bottle scope, a possible missing bottle annotation, small cylindrical drink-package class ambiguity, dense buffet scale/serving-granularity problems, and sealed food packaging detected as food. Actual material, contents and bin acceptance cannot be settled from these previews. No labels or sorting rules were changed.

A separate, previously completed bottle-specialist experiment tested one additional epoch of classification-head training. Validation AP50 declined from 82.47% to 77.95%, and AP50–95 fell from 62.00% to 59.18%. Because the candidate failed the predefined validation criterion, it was rejected without being activated or evaluated on the historical test cohort. The frozen system reported here was not retrained during manuscript finalization.

8.0.1 Exploratory 44-photo public-source stress test (qualitative; not independent)

After the research freeze, the unchanged four detectors were run on 44 source-screened public photographs (nine bottle-source, six carton-source, 15 utensil-source, 12 hinged-container-source and two wrapper-source images).

There were 149 displayed predictions at the 0.25 confidence floor: 12 compost suggestions, 49 recycling suggestions and 88 manual-review outcomes. Three photos had no displayed detections. These are **prediction counts only**: there are no human-verified object boxes, original capture-day independence certificates, or school bin-eligibility labels, so recognition mAP, correct-bin accuracy and sorting precision are **unmeasured** on this cohort.

A preliminary image-level *source-hint* cross-check found at least one matching model class on 9/9 bottle-source, 5/6 carton-source, 12/15 utensil-source, 4/12 hinged-container-source and 0/2 wrapper-source photos. These fractions are **not recall**: source names are not exhaustive human object labels, and some images contain multiple target items. Visual inspection suggests repeated plate/hinged-container confusion on open foam food boxes and missing wrapper detections. A foil meal tray was detected as `drink_carton` at confidence 0.617, triggering an automatic recycling suggestion under the frozen user-selected policy. This is a plausible wrong-object automatic action; actual local bin eligibility cannot be concluded from the photo alone. A carton photo with three cartons yielded four carton boxes and another carton image yielded none. Crowded bottle images also exhibited overlapping boxes and cross-specialist labels.

These observations supply **qualitative failure-mode examples**, not evidence of a measured performance change or an authorization to retrain on a future holdout. Individual photo references and limitations are documented in the visual audit. These public images may be out-of-distribution relative to actual cafeteria disposal scenes, and original source rights require review before reproducing them in a publication.

9 Limitations and Future Evaluation

The main limitation of this study is that the reported benchmark images are development data, not a sealed holdout. As demonstrated in research on distribution shift and leakage, performance on reused or mismatched source data may not transfer to the intended deployment setting [11, 12]. Earlier inference, model selection, and related capture groups prevent an independent generalization claim. The object annotations were also AI-assisted, with limited evidence of human verification; in particular, the hinged-container class has only two validation objects and four historical-test objects. Food-serving boundaries and visually similar packaging introduce further annotation uncertainty. The existing audit reduces uncertainty about what data and model versions were used, but incomplete historical records mean that the absence of a detected duplicate cannot establish independence.

A second limitation concerns the distinction between an object class and a disposal decision. The current rule layer does not inspect whether a bottle or carton is empty, whether food is contaminated by packaging, or whether a specific material is accepted by the school's collection program. It cannot automatically assign landfill disposal, and requesting manual review is an abstention rather than a correct sorting outcome. The results therefore cannot quantify correct-bin accuracy or reductions in waste contamination. Latency, throughput, energy use, and user interaction during a real cafeteria deployment have also not been measured. Source-license and attribution records must be checked before any external image redistribution.

The next evaluation should use newly collected cafeteria photographs from capture sessions excluded from all training, threshold selection, and prior inspection. Inclusion criteria and capture-group membership should be fixed before model inference. Annotators who have not seen the model's predictions should mark each relevant object's bounding box and actual disposal eligibility using documented local collection rules and, when necessary, material and contents information unavailable from the photograph. Uncertain cases should be adjudicated or recorded as unscorable under a predefined protocol. On that sealed cohort, the unchanged system can then be evaluated for object detection, correct-bin precision and recall, prediction coverage, and manual-review burden, with uncertainty estimated across independent capture groups rather than treating multiple objects in one photograph as unrelated observations.

10 Conclusion

Four specialized YOLO11 detectors recognized seven categories of cafeteria items with 87.78% mAP50 and 67.24% mAP50–95 on an exposed development cohort. Choosing confidence thresholds separately by class increased measured object-detection precision to 97.03% on validation and 97.22% on the historical cohort, although fewer true objects were detected. The experiment shows that selective thresholds can make the system more conservative without retraining its detectors. It does not show that the system sends items to the correct bin. Establishing that claim requires new school photographs, independent human annotations, and verified local disposal labels evaluated without further model or threshold tuning.

11 Reproducibility and Data Availability

Maintained code/configuration/results are in Git. Images, checkpoints, historical experiments and the detailed 35 MB asset inventory remain external and intact. GitHub alone therefore cannot reproduce native inference without the external checkpoints and data. Publishing the source does not distribute the external checkpoints or image datasets.

- `configs/research-freeze.json`: checkpoint, policy, manifest and benchmark pins.
- `results/sorting-verification.json`: prior verified replay and detection metrics.
- `results/research-readiness.json`: this audit's compact findings and null bin metrics.
- `scripts/research_audit.py`: conservative inventory, pixel/perceptual screening, provenance indexing and tested error-diagnostic functions.
- External `artifacts/research-finalization-20261101/`: protocol, complete image and metadata inventory, read failures, error details, visual-review selection/notes, environment metadata and shareable statistical figure. Its summaries link hashes of detailed evidence; generated files are analysis artifacts, not new datasets.

From the repo root, with a fresh output path:

```
../yolo-cafeteria-research-assets/venv/bin/python -B scripts/research_audit.py --output /tmp/cafeteria-ass
../yolo-cafeteria-research-assets/venv/bin/python -B -m unittest discover -s tests -v
```

This audit does not run YOLO or train. Its dHash/source paths are defined in code; future added files may change inventory totals without changing the frozen models. The main replay/evaluation reproduction commands and frozen receipts are in sorting-evaluation.md. Existing outputs refuse overwrite. For actual independent bin metrics, use the existing `scripts/evaluate_sorting.py` with new independently human-verified truth and the unchanged frozen sorting policy; the required label format and matching convention are documented in that report.

12 References

[1] U.S. Environmental Protection Agency. *Frequent Questions on Recycling.* Official recycling guidance. Accessed November 2026.

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